Harshavardan N

AI/ML Engineer | Data Scientist | Software Engineer

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Summary

Computer Science undergraduate specializing in Artificial Intelligence, Machine Learning, and Data Science with research experience at the Centre of Neuro Informatics, VIT Chennai. Experienced in developing AI-powered applications, predictive analytics solutions, secure edge computing systems, and serverless cloud architectures through 10+ end-to-end projects across healthcare, cybersecurity, finance, agriculture, and IoT. Co-authored a published patent in AI-assisted neuromodulation and passionate about leveraging intelligent systems to solve real-world engineering challenges through scalable, data-driven solutions.

Work

Intern May 2025 - Jul 2025

CENTER OF NEUROINFORMATICS, VITCC | Chennai, India

- Contributed to the design and development of a Cortico-Thalamic Phase-Locked Transcranial Focused Ultrasound (tFUS) System, aimed at enhancing stroke rehabilitation for patients.
- Co-authored a patent for the tFUS system, contributing to its technical claims, novelty, and documentation by performing simulations and complex calculations on actual EEG datasets to validate results.

Skills

Programming

Python, Java, C++, C, Javascript, SQL

AI / Data Science

Machine Learning, Predictive Analytics, NLP, XGBoost, Feature Engineering, Scikit-learn, Pandas, NumPy

Cloud

AWS Lambda, AWS CDK, Step Functions, SNS

Backend

Flask, React.js, Node.js, Express, REST API, MongoDB, SQLite

Developer Tools

Git, Github, Linux, VS Code

Projects

Adaptive Neuro Edge Federated Framework

Python | Federated Learning | Edge AI | Distributed Systems | Privacy-Preserving AI

Designed a privacy-preserving federated learning framework in Python for intelligent edge devices, enabling decentralized AI model training while minimizing centralized data dependency for healthcare-inspired applications.

- Developed a distributed federated learning architecture for edge AI.
- Implemented secure decentralized model aggregation.

Indian Cybersecurity Threat Detection System

Python | ESP32 | Machine Learning | NLP | Streamlit | MongoDB | Edge Computing

Developed an AI-powered cybersecurity monitoring platform using ESP32 edge devices, machine learning, NLP, and real-time analytics for intelligent threat detection.

- Built distributed telemetry collection using ESP32.
- Developed adaptive ML-based anomaly detection achieving **95.1% True Positive Rate** while reducing false positives to **2.8%**.

Insurance Fraud Propensity Prediction

Python | XGBoost | Scikit-learn | Predictive Analytics | Feature Engineering | Data Science

Developed a cost-sensitive machine learning model using XGBoost to identify fraudulent insurance claims through advanced feature engineering and predictive analytics.

- Achieved **67.3% recall** using a cost-sensitive XGBoost model optimized through feature engineering.

Serverless Geospatial Analysis Framework

AWS Lambda | AWS CDK | Amazon S3 | AWS Step Functions | Amazon SNS | Python | Serverless Computing

Designed a scalable AWS serverless architecture for automated geospatial data processing using event-driven cloud services and Infrastructure as Code.

- Designed AWS Lambda-based processing pipelines.
- Automated raster data ingestion.
- Implemented Step Functions orchestration.
- Used AWS CDK for Infrastructure as Code.

PUF IoT Monitoring System

ESP32 | React.js | Node.js | Express.js | SQLite | PUF | IoT Security

Engineered a secure IoT monitoring platform using ESP32-based Physically Unclonable Functions (PUFs) with a full-stack dashboard for real-time hardware authentication and monitoring.

- Designed a custom three-stage Ring Oscillator.
- Generated hardware-specific cryptographic identities.

Additional Projects

• Supply Chain Prescriptive Analytics

• Quantum-Enhanced Hyperspectral Crop Pathogenesis Simulation

• Constraint-Based Problem Solver

• Cloudy URL Shortener

• Flix – Responsive Movie Streaming Website

• EmotAI – AI Mental Health Support Chatbot

Education

Vellore Institute of Technology May 2027
Bachelor of Technology Computer Science Engineering
CGPA: 8.64

Publications

Published Patent

- ***Cortico-Thalamic Phase-Locked Transcranial Focused Ultrasound (tFUS) System***
- **Published under Section 11A, May 2026**
- **Co-author**

Certifications

Oracle AI Foundation Associate Jun 2026

Oracle Gen AI Professional Jun 2026

Training/Courses

Prompt Engineering (VIT) Mar 2026

The Complete Full-Stack Web Development Bootcamp (Udemy) May 2026

100 Days of Code™:: The Complete Python Pro Bootcamp (Udemy) May 2026

AWS Cloud Practitioner (Completion in Progress)